[AN INTERACTIVE AR KOREAN FOOD CUSTOMIZATION SYSTEM FOR ENHANCED DINING EXPERIENCE AND DECISION SATISFACTION]

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LIST OF ABBREVIATIONS

FYP	**-**	**Final Year Project**
AR	**-**	**Augmented Reality**
UI	**-**	**User Interface**
UX	**-**	**User Experience**
3D	**-**	**Three-Dimensional**
2D	**-**	**Two-Dimensional**
F&B	**-**	**Food and Beverage**
SDLC	**-**	**System Development Life Cycle**
APK	**-**	**Android Package Kit**
RAM	**-**	**Random Access Memory**
CPU	**-**	**Central Processing Unit**
GPU	**-**	**Graphics Processing Unit**
API	**-**	**Application Programming Interface**

Chapter 1: INTRODUCTION

1.1 Introduction

The advancement of digital technology has significantly transformed the way users interact with services across various industries, including the food and beverage (F&B) sector. In recent years, the growing global popularity of Korean cuisine has led to an increasing demand for more engaging and informative dining experiences. Korean dishes are commonly characterized by multiple ingredients, customizable components, sauces, and beverage selections, requiring users to make decisions based on several factors prior to placing an order.

However, current menu systems, whether in physical or digital form, remain largely static and limited in their ability to provide a comprehensive representation of food items. Most menus rely on two-dimensional (2D) photographs and textual descriptions, which may not accurately reflect the final appearance, portion size, or ingredient combinations of a dish. As a result, users may experience uncertainty when making food choices, which can consequently lead to dissatisfaction or ordering regret.

With the emergence of Augmented Reality (AR) technology, new opportunities have arisen to enhance user interaction and visualization. AR allows digital objects to be overlaid onto the real-world environment, enabling users to interact with virtual content in a more immersive and realistic manner. In the context of food ordering, AR can provide a three-dimensional (3D) visualization of dishes, allowing users to better understand their choices before making a decision.

Despite the potential of AR, most existing applications in the food domain focus primarily on visualization rather than interaction. Users are typically limited to viewing 3D models without the ability to customize ingredients, sauces, drinks, or receive real-time feedback on pricing and calorie information. This limitation reduces the overall effectiveness of AR as a decision-support tool in food ordering contexts.

Therefore, this project proposes the development of an interactive AR-based Korean food customization system that enables users to visualize and modify food components in real time. The system aims to enhance user experience, improve decision-making confidence, and increase overall satisfaction throughout the dining process.

1.2 Problem Statement

The current food ordering experience faces several limitations that affect user decision-making and satisfaction. Firstly, traditional menu systems rely heavily on static images and textual descriptions, which are insufficient to represent complex and customizable food items. Users are unable to clearly visualize the final outcome of their selected ingredients, food variations, ingredient combinations, sauces, and drinks. This lack of clarity often leads to uncertainty during the decision-making process. Secondly, although Augmented Reality technology has been introduced in food-related applications, most implementations are limited to basic visualization features. These systems allow users to view 3D representations of food but do not support interactive customization. Essential features such as ingredient customization, sauce and drink selection, real-time food visualization, dynamic price calculation, and calorie information are often not included. Thirdly, there is a lack of comprehensive evaluation on how AR-based systems influence user decision-making, satisfaction, and engagement in the context of food customization. Existing studies primarily focus on user experience or product visualization without addressing the integration of interaction and decision support. Therefore, there is a need to develop a more advanced AR system that enables interactive food customization while improving user decision confidence, engagement, and satisfaction.

1.3 Objectives

I. To study the decision-making challenges and satisfaction levels associated with conventional food menu systems through literature review and preliminary user research.

II. To design and develop an interactive AR-based Korean food customization system that incorporates ingredient selection, sauce and drink customization, and real-time price and calorie calculation using Unity and AR Foundation.

III. To evaluate the developed system in terms of user satisfaction, engagement, and decision confidence through structured user testing.

1.4 Scope

This project focuses on the development of a mobile-based Augmented Reality application for Korean food customization within a defined scope. The system will be developed using Unity Engine with AR Foundation, specifically targeting Android devices that support ARCore. The application will include four Korean food menus, namely Bibimbap, Ramen, Fried Rice, and Corndog dish that allows users to customize ingredients, sauces, drinks, and view real-time changes in visual representation, price, and calorie information. In addition, the system will include basic interaction features such as ingredient selection and user interface controls to enhance usability. User testing will be conducted using a structured questionnaire to evaluate the effectiveness of the system in terms of user satisfaction, engagement, and decision confidence. However, the project does not include integration with real restaurant systems, payment processing functionality, or large-scale deployment. The development is limited to a prototype level due to time constraints and the focus on core AR interaction design and evaluation.

1.5 Project Significant

This project offers several significant contributions to both users and the research field. For users, the system enhances the food ordering experience by providing an interactive platform that allows them to visualize and customize food in real time. This reduces uncertainty, improves decision confidence, and increases overall satisfaction while providing immediate feedback on food pricing and calorie information. For developers and researchers, this project contributes to the exploration of AR interaction design in consumer-oriented applications, particularly within the food and beverage domain. It also provides insights into how AR-based systems can influence user engagement and decision-making behaviour. Furthermore, the project bridges the gap between static visualization and interactive customization, offering a more comprehensive approach to improving user experience in digital menu systems.

1.6 Summary

This chapter has presented an overview of the project, including the background, problem statements, objectives, scope, and significance. The limitations of existing menu systems and AR applications have been identified, highlighting the need for an interactive AR-based food customization system. The proposed system aims to enhance user experience by enabling real-time visualization and customization, while also improving decision-making confidence and satisfaction. The next chapter will discuss the related literature and studies that support the development of this project.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

This chapter reviews existing literature relevant to the development of K-CustomAR, an interactive Augmented Reality-based Korean food customisation system. The review is organised around six themes: the fundamentals of AR technology, AR tracking methods, the application of AR in the food industry, user experience and interaction design in AR systems, the influence of AR on consumer decision-making, and the relationship between AR and customer satisfaction and engagement. Together, these themes build the theoretical and empirical foundation that justifies the proposed system and guides its design, development, and evaluation.

A comparison of fifteen existing studies is also presented to identify specific gaps in current implementations. All reviewed studies were selected based on their relevance to the project objectives and their contribution to understanding AR interaction, user-centred design, and decision satisfaction in consumer-facing applications.

2.2 Existing System And Related Studies

2.2.1 Augmented Reality Technology Overview

Before building any AR application, it helps to understand where the technology currently stands. Two broad survey studies provide that foundation.

Mendoza-Ramírez et al. (2023) conducted a systematic literature review to provide a comprehensive understanding of augmented reality technology, its applications, and its limitations across multiple domains. The study highlights that AR enhances real-world environments by overlaying digital content, thereby improving user interaction, perception, and decision-making. The authors classify AR into distinct types, namely marker-based, markerless, projection-based, superimposition-based, and outlining AR, and discuss their respective applications in sectors including education, healthcare, and retail.

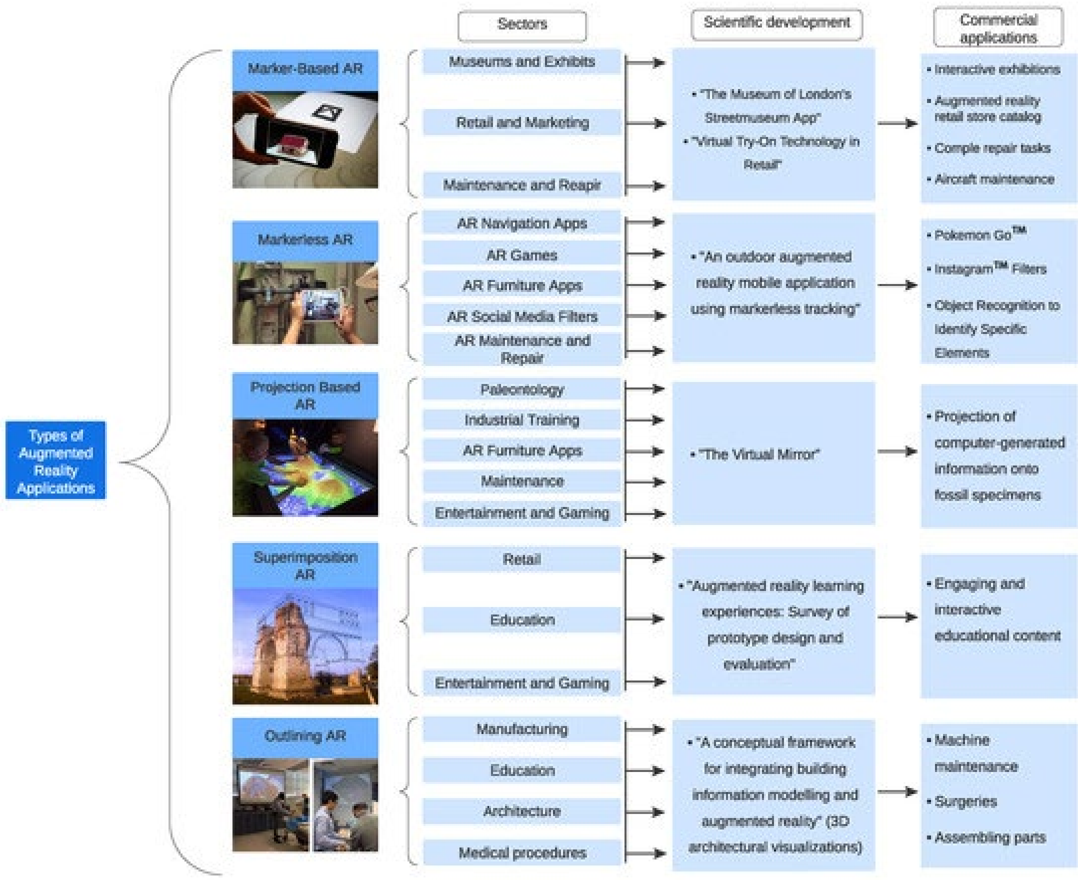

Figure 2.1 AR types classification diagram

Despite providing a broad theoretical foundation, the study remains general in scope and does not address domain-specific implementations such as food customisation or user-centred evaluation of decision satisfaction. This gap reinforces the need to develop specialised and interactive AR systems for targeted contexts.

Cao et al. (2023) narrowed the lens to mobile AR specifically, analysing 37 development frameworks and tracing the full technical pipeline from camera capture through machine learning processing to real-time rendering. Their survey spans education, healthcare, and marketing, and is particularly useful for understanding the infrastructure behind any mobile AR application. Like Mendoza-Ramirez et al., however, it does not explore food customisation as a use case, and user decision-making processes go unexamined. Both studies confirm what AR can do in broad terms; this project examines what it can do in one specific, practical context..

2.2.2 AR Tracking Methods: Marker-Based and Markerless

The choice of tracking method is one of the most consequential technical decisions in any AR system. Sulistiyono et al. (2024) tackled this directly by comparing marker-based and markerless tracking across 72 test combinations covering different distances, angles, and lighting conditions on two Android devices. The results showed that markerless tracking achieved a success rate of 94.4%, while marker-based tracking ranged from 72.2% to 77.8% depending on the device. Although markerless tracking offers greater flexibility in certain environments, marker-based image tracking remains suitable for applications that rely on predefined visual targets. Since K-CustomAR uses menu images as tracking markers, image-based tracking provides reliable object recognition, stable AR visualisation, and a consistent user experience during food customisation.

For K-CustomAR, these findings provide useful insights into the strengths and limitations of different AR tracking approaches. Since the system is designed around predefined menu images, marker-based image tracking was selected as it offers accurate target recognition and stable AR content rendering while maintaining ease of implementation for food visualisation and customization.

2.2.3 Augmented Reality in the Food Industry

The food and beverage industry has attracted a growing body of AR research, and a consistent pattern runs through most of it: the technology is used for visualisation, but rarely for interaction.

Chai et al. (2022) analysed 103 studies on AR and mixed reality in food contexts and found that AR reliably improves food-related experiences. It makes dishes look more appealing, communicates nutritional information at a glance, and can even prime how flavours are anticipated. AR has also been shown to influence food perception by altering the visual appearance of a dish in real time. Despite this, the review found that most applications stop at the visualisation layer. Interactive customisation barely features, and user-centred evaluation that measures decision satisfaction and engagement is largely absent. This gap is central to what K-CustomAR sets out to address.

Amin et al. (2023) presented an AR-based interactive food menu system developed for Android, which overlays three-dimensional food models onto QR code image markers using the Vuforia SDK integrated with Unity 3D.

Figure 2.2 Screenshot of AR food menu system on phone

The system also displays ingredient information, nutritional content, and pricing when the marker is scanned. The study demonstrates a practical implementation of AR in a restaurant context and highlights the potential of mobile AR to improve customer satisfaction and provide a more informative ordering experience. However, the system offers limited interactive customisation and does not evaluate the impact on user decision confidence. This gap provides justification for extending AR food systems to include deeper customisation features and user satisfaction evaluation.

Ismail and Fadzli (2025) attempted to go further by combining AR and VR into a single extended reality (XR) food ordering system, recognising that neither technology alone was sufficient. AR without customisation limits user engagement; VR in isolation risks causing motion sickness due to scene complexity. Their XR-

Menu system is technically ambitious, but the study stops before a full user evaluation is completed. It confirms the direction of travel without providing evidence of outcome.

Kontogianni and Bouki (2021) reviewed the current state of AR applications specifically in food promotion and food analysis, identifying 34 indicative AR applications across various purposes and implementations. The study classifies applications using eight criteria including content, context, execution scenario, device support, and markers used. The findings highlight the growing use of AR for food promotion, interactive product experiences, and nutritional awareness. However, most reviewed applications serve promotional or informational purposes rather than supporting user-driven interactive customisation and decision-making. This confirms the need for more comprehensive AR food systems that integrate customisation and user evaluation.

Scott et al. (2023) conducted an empirical study to investigate how AR influences food desirability and purchase intention through field and laboratory experiments. The findings revealed that AR significantly increases food desirability and purchase likelihood, primarily through mental simulation, whereby users imagine consuming the food when they see it overlaid in their real environment. AR also increases the perceived personal relevance of food, further strengthening the decision-making process. However, the study focuses on visualisation only and does not incorporate interactive food customisation or system-based implementation. There is therefore an opportunity to develop an interactive AR food system that enhances user decision satisfaction through customisation features.

2.2.4 User Experience and Interaction Design in AR

User experience is not a secondary concern in AR development, it is often the difference between a system that gets adopted and one that does not. Two studies offer important insight into how UX should be understood and measured in mobile AR contexts.

Dirin and Laine (2018) investigated user experience in mobile augmented reality applications using a mixed-method approach including case studies, Likert-scale questionnaires, interviews, and observations. The findings indicate that MAR applications create strong emotional engagement, with users reporting feelings of excitement, inspiration, and enjoyment.

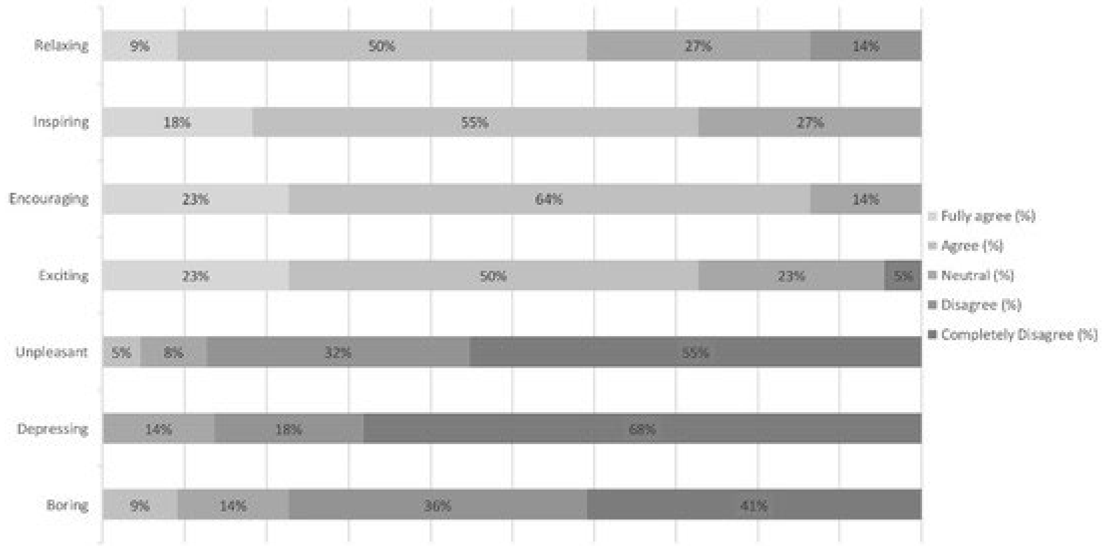

Figure 2.2 UX evaluation framework or emotion scale diagram

The study also highlights that user experience plays a critical role in the success and acceptance of AR systems, and identifies challenges including performance issues, complexity, and usability limitations. Although the study provides valuable UX design insights, it does not focus on specific domains or decision-making processes, creating an opportunity to apply these evaluation methods in a food customisation AR context.

Building on this, Picardi and Caruso (2024) proposed a user-centred evaluation framework for AR applications by analysing evaluation methods across 399 studies. The study highlights that many AR applications focus primarily on technical development while neglecting user experience and usability. The framework categorises evaluation methods such as questionnaires, observations, interviews, and user action logging, with questionnaires being the most widely used method at 36% of studies.

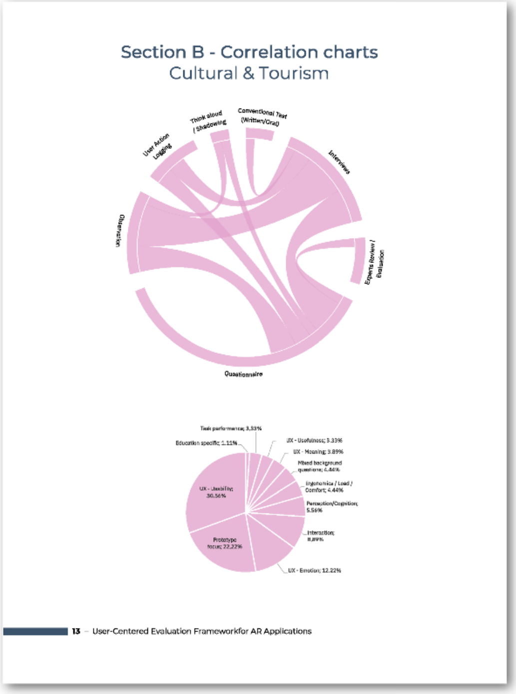

Figure 2.3 Evaluation framework diagram

Key UX components identified include emotion, usability, usefulness, and meaning. However, the framework is general and does not target specific domains such as food customisation, reinforcing the need to apply it to specialised AR systems for domain-specific evaluation of user satisfaction and decision-making.

2.2.5 AR and Consumer Decision-Making

The influence of AR on consumer decision-making is a central concern for this project. Several studies have examined how AR technology affects decision confidence, purchase intention, and user behaviour.

Qin et al. (2021) surveyed 162 participants using a model grounded in Stimulus-Organism-Response (S-O-R) theory to examine how mobile AR shapes shopping behaviour. The findings were clear: AR reduces uncertainty, improves product visualisation, and increases user confidence in purchase decisions. Critically, the effect was strongest when AR was perceived as both enjoyable and useful, also hedonic and utilitarian value working together. When users found the system fun to use and practically informative, they were significantly more likely to make confident, positive decisions. This finding directly informs the design priorities of K-CustomAR:

the system must be engaging as well as informative. The limitation is that the study focused on retail products, not food, which leaves an empirical gap that this project fills.

Hilken et al. (2020) explored the role of social augmented reality in supporting shared decision-making through multiple experiments analysing recommender-decision maker interactions. The findings indicate that AR facilitates collaboration by enabling users to share a common visual perspective and provide recommendations in real time, supported by the concept of social empowerment. Image-enhanced communication was found to be more effective than text-based recommendations.

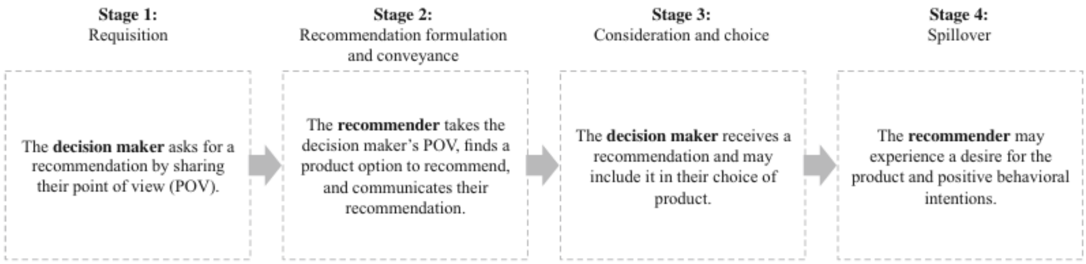

Figure 2.4 4-stage decision process model diagram

However, the study focuses on social interaction rather than individual decision-making and does not include interactive customisation. There is therefore an opportunity to develop AR systems that support individual decision-making through personalised and interactive food customisation experiences.

2.2.6 Customer Satisfaction and Engagement with AR

Beyond decision-making, a consistent thread across the literature is that AR improves how satisfied and engaged users feel and outcomes that are directly evaluated in this project.

Poushneh and Vasquez-Parraga (2017) measured the impact of AR on retail customer experience through a lab experiment assessing pragmatic quality, aesthetic quality, and hedonic quality. AR significantly enhanced all three dimensions, leading to higher customer satisfaction and stronger purchase intention. The study is retail-focused, but its UX measurement framework translates well to the food customisation context. It supports the expectation that K-CustomAR should produce measurable satisfaction gains relative to conventional menu systems.

Fan et al. (2020) approached the same question from a cognitive angle. Using a 2x2x2 experimental design, they found that AR reduces cognitive load and increases cognitive fluency, allowing users to process product information more efficiently. When AR supported environmental embedding and simulated physical control and letting users feel they were interacting with the product rather than just viewing it. So product attitudes became more positive and engagement increased. This is a useful framing for K-CustomAR's toggle-based interaction design: ingredient selection is not just a functional feature; it is a mechanism for cognitive engagement.

Alyahya et al. (2024), studying AR and VR adoption in the hospitality sector, found that perceived ease of use, innovativeness, and usefulness all positively influence satisfaction and intention to continue using the technology. Importantly, they also found that AR reduces uncertainty and improves confidence in service-related decisions and an effect observed in the hotel industry but consistent with what Qin et al. (2021) found in retail. Both studies point in the same direction: when AR is well-designed and easy to use, it makes people more confident and more satisfied. K-CustomAR is built on exactly this premise.

2.3 Comparison of Existing Systems

The following table summarises and compares the fifteen reviewed studies in terms of author and year, article title, methodology employed, key findings, and relevance to this project.

Table 2.1: Comparison of Existing Studies

No.	Author & Year	Article Title	Methodology	Key Findings	Relevance to Project
1	Mendoza-Ramírez et al. (2023)	Augmented Reality: Survey	Systematic literature review; ~200 papers from Scopus.	Reviews AR types and applications; AR improves interaction and decision-making.	Provides foundational AR knowledge and classification for system development.

2	Cao et al. (2023)	Mobile Augmented Reality: User Interfaces, Frameworks, and Intelligence	Systematic survey of 37 MAR frameworks.	Identifies MAR frameworks, UI design challenges, and the AR system pipeline.	Supports design of AR interface and system architecture.
3	Sulistiyono et al. (2024)	Comparative Study of Marker-Based and Markerless Tracking in AR	MDLC methodology; 72 test scenarios across two devices.	Markerless AR achieved 94.4% success rate vs 72.2–77.8% for marker-based.	Informs AR tracking method selection; supports markerless approach.
4	Chai et al. (2022)	Augmented/Mixed Reality Technologies for Food: A Review	Systematic review of 103 food AR/MR studies.	AR improves food visualisation and influences food perception and decision-making.	Directly supports AR food visualisation concept.
5	Amin et al. (2023)	An AR-Based Approach for Designing Interactive Food Menu Using Android	System development using Unity 3D and Vuforia SDK.	AR food menu overlays 3D model with ingredient and pricing info.	Most similar existing system; highlights need for interactive customisation.
6	Ismail & Fadzli (2025)	XR-Menu: Food Ordering System in Extended Reality	System development; XR integration of AR and VR.	XR improves food ordering; AR and VR alone have limitations.	Supports concept of advanced interactive food ordering using AR.
7	Kontogianni & Bouki (2021)	Augmented Reality in Food Promotion and Analysis	Literature review of 34 AR food applications using 8 criteria.	AR mainly used for food promotion; limited interactive customisation.	Confirms gap in interactive AR food customisation.
8	Scott et al. (2023)	From Tablet to Table: How AR Influences Food Desirability	4 experimental studies; AR vs non-AR food presentation.	AR increases food desirability through mental simulation.	Supports AR's role in improving food decision satisfaction.
9	Dirin & Laine (2018)	User Experience in Mobile Augmented Reality	Mixed methods: case studies, Likert questionnaires, interviews.	MAR creates positive emotional engagement; UX critical for AR acceptance.	Supports evaluation of user satisfaction and engagement.

10	Evangelista et al. (2024)	User-Centered Evaluation Framework for AR Applications	Framework from analysis of 399 studies; cluster analysis.	Questionnaire most used method; UX includes emotion, usability, usefulness.	Guides user testing evaluation design for this project.
11	Qin et al. (2021)	How Mobile AR Influences Consumer Decision Making	Quantitative survey using S-O-R model; 162 participants.	AR reduces uncertainty and improves decision quality and confidence.	Core support for decision confidence objective.
12	Hilken et al. (2020)	Seeing Eye to Eye: Social AR and Shared Decision Making	5 experimental studies; Likert-based measurement.	AR supports shared decision-making through social empowerment.	Supports decision-making theory for this project.
13	Poushneh & Vasquez-Parraga (2017)	Discernible Impact of AR on Retail Customer Experience	Lab experiment; Likert questionnaire; pragmatic and hedonic UX dimensions.	AR improves UX, leading to higher satisfaction and purchase intention.	Supports user satisfaction and engagement objectives.
14	Fan et al. (2020)	Adoption of AR in Online Retailing and Consumers' Product Attitude	2x2x2 experimental design; Likert questionnaire.	AR reduces cognitive load and increases cognitive fluency.	Supports engagement and cognitive processing aspects.
15	Alyahya et al. (2024)	Augmented and Virtual Reality in Hotels: Impact on Tourist Satisfaction	Quantitative study; Likert questionnaire with hotel customers.	AR ease of use and usefulness positively influence satisfaction.	Confirms AR enhances satisfaction and decision confidence in service contexts.

Based on the comparison above, several common gaps can be identified across the reviewed studies. Most existing systems focus primarily on AR visualisation without providing interactive customisation features such as ingredient selection or real-time price calculation. Furthermore, none of the reviewed studies propose a system specifically designed for Korean food customisation in a mobile AR environment. User-centred evaluation of decision confidence and satisfaction also remains limited in food-specific AR contexts. These gaps collectively justify the development of the proposed interactive AR Korean food customisation system in this project.

2.4 Summary

This chapter has reviewed fifteen studies across six thematic areas which are AR technology, tracking methods, AR in the food industry, user experience and interaction design, consumer decision-making, and customer satisfaction and engagement. The literature consistently confirms that AR is effective at improving user interaction, supporting better visualisation, and increasing both decision confidence and overall satisfaction in consumer-facing contexts.

What the literature also reveals, however, is a persistent gap between what AR can do and what most implementations deliver. Systems tend to be built for display rather than interaction. Food-specific AR applications rarely incorporate customisation features, and user-centred evaluation of decision confidence in food ordering contexts is limited. K-CustomAR is designed to address these gaps directly and combining interactive AR customisation with structured evaluation of the user experience outcomes that matter most. Chapter 3 will present the project methodology in full, covering the development approach, requirement analysis, tools and technologies, and the evaluation plan.

CHAPTER 3: METHODOLOGY

3.1 Introduction

This chapter explains how K-CustomAR was built and how its effectiveness will be measured. It covers the development methodology chosen for the project, the process used to gather and define system requirements, the tools and platforms selected for development, and the plan for testing with real users. Every decision described here connects directly to the three objectives set out in Chapter 1 which is to build a system that is not only technically sound but genuinely useful and satisfying for its intended users.

3.2 Project Methodology

K-CustomAR is developed using the System Development Life Cycle (SDLC), a well-established framework that moves a project through five sequential phases: analysis, design, development, implementation, and evaluation. The SDLC was selected because it suits the nature of this project well — requirements are clearly defined, the feature set is bounded, and the evaluation has a fixed endpoint. Each phase must be completed and validated before the next begins, which reduces the risk of late-stage issues and keeps the work aligned with the project timeline. As Sommerville (2016) notes, the SDLC is particularly appropriate for software projects where the scope and requirements can be determined upfront. The five phases as applied to K-CustomAR are described below.

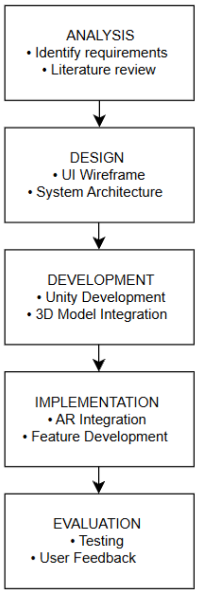

Figure 3.1System Development Life Cycle (SDLC) methodology used in the development of K-CustomAR.

3.2.1 Analysis

The analysis phase began with a structured review of existing food menu systems and AR applications to identify where current solutions fall short. This review, presented in Chapter 2, revealed two clear problems: traditional menus cannot meaningfully represent customisable food items, and most AR food applications limit users to passive viewing with no interactive customisation. These findings were used to define the core problem the system must solve and to establish the functional requirements that would guide everything that followed. At this stage, the key question was not what AR can do in general, but what it specifically needs to do to improve the food ordering experience.

3.2.2 Design

The design phase focuses on planning the system architecture, user interface, and interaction flow of the K-CustomAR application. The application is designed to provide users with an interactive Augmented Reality experience for visualising and customising Korean food items before making ordering decisions.

The system workflow begins when users scan an AR menu marker using their mobile device. Once the marker is successfully detected, the application displays a home menu containing several Korean food categories, namely Bibimbap, Ramen, Fried Rice, and Corndog. Users can select a food category and proceed to the corresponding customisation interface.

The user interface is designed to be simple and intuitive, allowing users to customise food items through interactive buttons. Depending on the selected menu, users can modify ingredients, choose sauces, add side items, and view real-time updates of food prices and calorie information. An order summary interface is also designed to display the user's final customisation selections before completing the session.

To ensure a consistent user experience, the interface layout follows user-centred design principles, emphasising clarity, ease of navigation, and visual feedback. Storyboards, wireframes, and user flow diagrams are developed during this phase to illustrate the interaction process and screen transitions within the application.

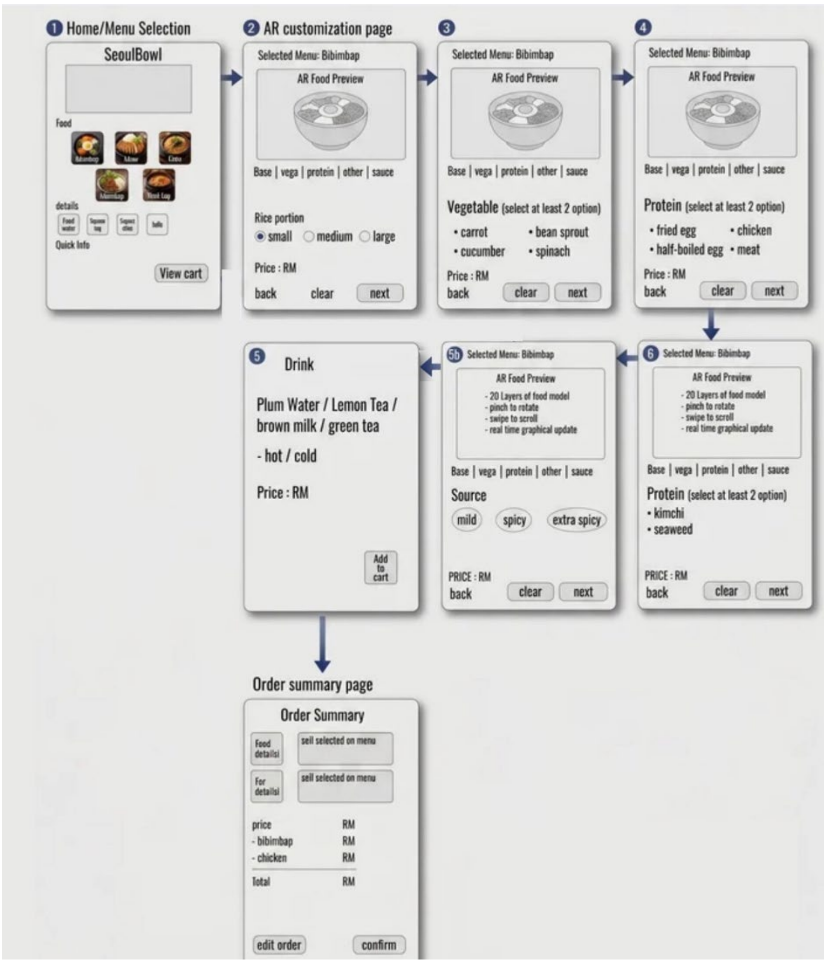

Figure 3.2 Initial wireframe design of K-CustomAR interface

3.2.3 Development

The development phase involved constructing the AR application using the Unity game engine integrated with AR Foundation. Three-dimensional food models were created in Blender and imported into Unity for rendering. All interaction features like ingredient selection, sauce and drink selection, real-time price and calorie calculation, and order summary generation were implemented using C# scripting within Unity. Performance logging scripts were also embedded in the project to collect technical data automatically during testing sessions. The application is built to run on Android devices that support ARCore.

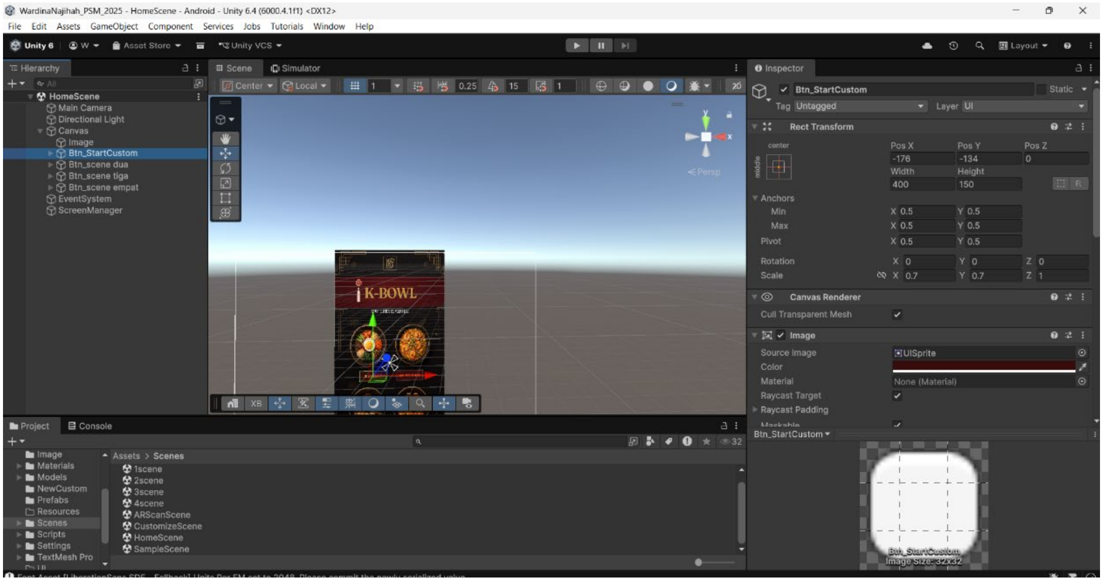

Figure 3.3 Development environment in Unity showing AR food customization components

3.2.4 Implementation

During the implementation phase, the completed application was deployed to Android devices and made available to test participants. A minimum of 30 participants interacted with the system and completed the evaluation instruments after each session. Technical performance data was collected automatically in the background through the embedded logging scripts, requiring no manual intervention during testing.

3.2.5 Evaluation

The evaluation phase assessed K-CustomAR through two complementary approaches: subjective survey-based testing and objective technical performance testing. Survey instruments were administered to participants after each testing session to capture their experience with the system. Technical performance was assessed through Unity-based logging scripts that recorded system metrics throughout each session. Together, these two approaches provided a complete picture of how well the system met its intended objectives.

3.3 Requirement Analysis

3.3.1 Need Analysis

The development of K-CustomAR is motivated by the limitations of existing food menu systems, which rely on static 2D images and text descriptions that are insufficient for representing customisable food items. Users of Korean food restaurants, particularly younger consumers aged 18 to 35, require a more informative and engaging ordering experience. AR technology provides an opportunity to bridge this gap by enabling real-time 3D food visualisation and interactive customisation. The system is developed to fulfil the following functional requirements:

• Allow users to scan a menu marker and view a 3D Korean food model in real time.

• Allow users to customise Korean food items by selecting ingredients, sauces, and side options.

• Allow users to view an order summary displaying selected menu items, sauces, drinks, total price, and calorie information before completing the ordering session.

• Display real-time price and calorie updates based on the selected customisation options.

• Generate an order summary showing the final food configuration chosen by the user.

• Provide an interactive and engaging AR-based food visualisation experience to support food ordering decisions.

3.3.2 User Analysis

The target users of K-CustomAR are consumers interested in Korean cuisine, particularly individuals aged 18 to 35 who are comfortable using mobile applications and digital interfaces. Users are assumed to own Android smartphones capable of supporting ARCore. Based on the literature reviewed in Chapter 2, users in this demographic respond positively to interactive, visually engaging digital experiences and are more likely to adopt AR tools that reduce the effort required to make decisions. The interface is designed to be intuitive and accessible regardless of the user's level of technical experience.

3.3.3 Technical Analysis

K-CustomAR is developed using Unity 6 (Version 6000.4.1f1) integrated with AR Foundation and the ARCore XR Plugin for Android AR functionality. Three-dimensional food models are created in Blender 3.6. The application targets Android devices running Android 8.0 (API Level 26) or higher with ARCore support. The minimum hardware requirement for the target device is 3 GB RAM with an ARCore-certified chipset.

3.3.4 Requirement Gathering

Requirements were gathered through the review of existing AR food systems and related literature presented in Chapter 2. Studying systems such as Amin et al. (2023) and Ismail and Fadzli (2025) highlighted what current applications offer and, more importantly, where they fall short. The most consistent gap was the absence of interactive customisation and structured user evaluation. These observations were documented and used to define the system scope, shape the interaction design, and determine the evaluation criteria for this project.

3.4 Project Requirements

3.4.1 Software Requirements

The following software tools are required for the development of K-CustomAR:

- Unity 6 (Version 6000.4.1f1) -primary development environment for AR application.
- AR Foundation (Unity package) -AR session management and cross-platform AR support.
- ARCore XR Plugin -Android ARCore integration within Unity.
- Blender 3.6 -3D food model creation and UV unwrapping.
- Visual Studio 2022 -C# scripting for AR interaction features and logging.
- Android Studio / SDK -Android build tools and device debugging.
- Google Forms -digital questionnaire administration for SUS and UEQ surveys.
- Microsoft Excel / SPSS -data analysis for survey results and performance metrics.

3.4.2 Hardware Requirements

The following hardware is required for development and testing:

- **Development Laptop:** AMD Ryzen 5 7535HS processor (3.30 GHz), 16 GB RAM, 4 GB dedicated graphics card, and 477 GB SSD storage for Unity development, AR implementation, and 3D asset integration.
- **Android Tier 1 (Entry-level):** 3–4 GB RAM, ARCore-supported chipset (e.g., Snapdragon 460).
- **Android Tier 2 (Mid-range):** 6 GB RAM, ARCore-supported chipset (e.g., Snapdragon 680).
- **Android Tier 3 (High-end):** 8 GB or higher RAM, ARCore-supported chipset (e.g., Snapdragon 8 Gen 1).
- **USB Cable:** Used for Android device debugging, testing, and APK deployment.

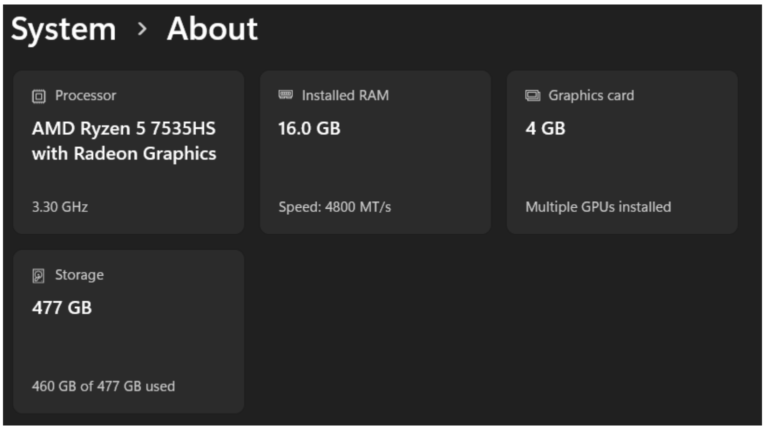

Figure 3.4 Development Laptop Specifications

3.4.3 Other Requirements

Additional resources required include a quiet testing environment to ensure consistent AR tracking conditions during user sessions. Printed menu cards with AR image targets are needed as physical markers for the scanning feature. A Google Forms link will be distributed digitally to participants following each session. Computer lab access may be used for data analysis and report writing.

Figure 3.5 AR image target used for food visualization and customization

3.5 Testing and Evaluation Plan

The evaluation of K-CustomAR uses a dual-method approach that combines subjective survey-based testing with objective technical performance testing. Both types of evaluation are conducted during PSM 2.

3.5.1 Survey-Based Testing (Subjective)

A minimum of 30 participants will be recruited for user testing, in accordance with the Central Limit Theorem which states that a sample of $n \geq 30$ approaches a normal distribution and supports valid statistical analysis (Sulaiman, 2025/2026). The following standardised survey instruments will be administered:

Table 3.1 Survey Instruments for K-CustomAR Evaluation

Instrument	What It Measures	Items / Scale	Reference
SUS (System Usability Scale)	Usability of the AR application interface	10 items; score 0–100 (≥68 = acceptable)	Brooke (1996)
UEQ (User Experience Questionnaire)	Overall user experience across 6 dimensions: attractiveness, perspicuity, efficiency, dependability, stimulation, novelty	26 items; 7-point semantic differential scale	Laugwitz et al. (2008)
Satisfaction Questionnaire	User decision confidence, engagement, and satisfaction during food customisation	10–15 items; 5-point Likert scale	Adapted from Qin et al. (2021); Poushneh & Vasquez-Parraga (2017)

Survey data will be collected using Google Forms and exported to Microsoft Excel for descriptive analysis (mean, standard deviation) and SPSS for inferential analysis where applicable. SUS scores will be interpreted using the standard SUS grading scale, and UEQ results will be benchmarked against the UEQ dataset.

3.5.2 Technical Performance Testing (Objective)

Technical testing is conducted to evaluate the system's performance under real operating conditions across three tiers of Android devices. A minimum of three technical metrics will be measured as listed in Table 3.2.

Table 3.2 Technical Performance Metrics for K-CustomAR

Metric	Description	Data Collection Method	Target Benchmark
FPS & Frame Time	Frame rate and rendering smoothness during food model customisation	PerformanceLogger.cs (CSV log)	≥30 FPS; frame time ≤33ms
Model Load / Instantiate Time (s)	Time taken to load and display each food ingredient model	Timestamp before/after Instantiate() in C#	As low as possible; consistent across devices
Memory Usage (MB)	RAM consumption as number of food items increases during customisation	PerformanceLogger.cs (CSV log)	Stable; no sudden spikes

Performance data is collected automatically using the PerformanceLogger.cs script embedded in the Unity project, which logs FPS, frame time, and memory usage to a CSV file every second. The CSV file is retrieved from the device after each testing session and processed in Microsoft Excel to generate performance graphs. Testing is conducted across three Android device tiers (entry, mid-range, and high-end) to evaluate performance consistency across hardware configurations.

3.5.3 Hardware Configuration Matrix

Performance testing will be conducted across three Android device tiers to ensure the application performs adequately on a range of devices used by target users. Table 3.3 presents the hardware configuration matrix for K-CustomAR testing.

Table 3.3 Hardware Configuration Matrix for Performance Testing

Device Tier	Specifications	RAM	Android Version	ARCore Support
Tier 1 (Entry-Level)	Android smartphone	3–4 GB	Android 10 or above	ARCore certified
Tier 2 (Mid-Range)	Android smartphone	6 GB	Android 11 or above	ARCore certified
Tier 3 (High-End)	Android smartphone	8 GB+	Android 12 or above	ARCore certified

The metrics of FPS, model load time, and memory usage will be compared across all three device tiers. The expected output graphs include a line graph of FPS during customization, a bar chart of model load time per food item, and a line graph of memory usage as the number of food items increases.

3.5.4 Functional and AR Tracking Testing

In addition to survey-based evaluation and technical performance testing, functional testing and AR tracking testing will be conducted to verify that all system features operate correctly. Functional testing focuses on validating the core features of K-CustomAR, including AR marker scanning, scene navigation, food customization, ingredient selection, drink selection, real-time price calculation, calorie calculation, and order summary generation.

AR tracking testing is conducted to evaluate the reliability of image target detection under different operating conditions. The AR marker will be tested at different viewing distances, viewing angles, and lighting conditions to ensure stable tracking performance. Detection time will be recorded using the ARDetectionLogger script integrated within the Unity project. The testing results will be used to identify tracking issues and verify that the AR functionality operates consistently on supported Android devices.

Table 3.4 Functional and AR Tracking Test Cases

Test Area	Test Case	Expected Result
Marker Detection	Scan AR marker	Marker detected successfully
Scene Navigation	Navigate to Home Scene	Correct scene loaded
Food Selection	Select food menu	Corresponding scene displayed
Ingredient Selection	Toggle ingredients	Food model updated correctly
Drink Selection	Select drink	Order information updated

Price Calculation	Add/remove ingredients	Price updated correctly
Calorie Calculation	Add/remove ingredients	Calories updated correctly
Order Summary	Generate summary	Correct order details displayed

3.6 Project Schedule and Milestones

The project is carried out across two semesters. PSM 1 (January–June 2026) focuses on planning, literature review, requirement analysis, and system design. PSM 2 (July–September 2026) focuses on system development, implementation, testing, and report finalisation. Table 3.4 presents the project schedule and milestones.

Table 3.5 Project Schedule and Milestones

Activity	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14	W15
Literature Review															
Requirements Analysis															
System & UX Design															
3D Asset Creation															
AR Development (Unity)															
Integration & Unit Testing															
User Testing															
Data Analysis															
Draft Report Writing															
PSM 1 Presentation															

3.7 Summary

This chapter has presented the full methodology adopted for K-CustomAR. The SDLC framework provides a clear, structured path from initial analysis through to evaluation, and each of its five phases has been described in relation to the specific activities of this project. The requirement analysis established who the system is for, what it needs to do, and what tools are required to build it. The testing and evaluation plan covers both the subjective experience of users and the objective technical performance of the system, using standardised instruments and automated logging across three Android device tiers. The project schedule sets out the timeline for completing all activities across PSM 1 and PSM 2. Chapter 4 will present the full implementation of the system, detailing how each component was built, integrated, and prepared for deployment.

CHAPTER 4: IMPLEMENTATION

4.1 Introduction

This chapter presents the implementation phase of K-CustomAR, an interactive Augmented Reality-based Korean food customisation system developed for Android mobile devices. Implementation is where the design and requirements from Chapter 3 become a working application. This chapter covers the system architecture, user interface design, media creation, media integration, and product configuration management.

The end goal of this phase is a fully functional AR mobile application. When complete, a user should be able to scan a physical menu marker with their Android device, arrive at the Home Scene, choose one of four Korean food categories which are Bibimbap, Ramen, Fried Rice, or Corndog and customise their meal through interactive controls, watch the price and calorie count update in real time, and confirm their order through a summary screen. The application is developed using Unity 6 (version 6000.4.1f1), AR Foundation, ARCore, and C# scripting, and is delivered as an Android APK file.

4.2 System Architecture

K-CustomAR follows a client-side, single-device architecture in which all processing, 3D rendering, and application logic are executed locally on the user's Android device. No internet connection or backend server is required, as all food data, 3D models, pricing, and calorie values are embedded within the Unity application build. Figure 4.1 illustrates the overall system architecture of K-CustomAR, showing the sequential flow from user interaction through AR processing to output generation.

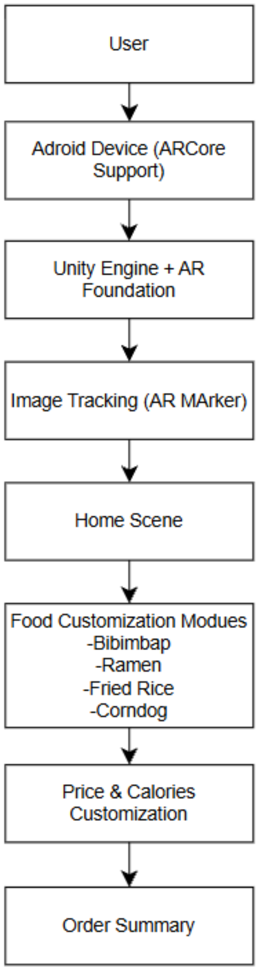

Figure 4.1 K-CustomAR System Architecture Diagram

The architecture follows a top-down sequential flow. The user initiates the session by pointing the device camera at the physical AR menu marker. The camera feed is processed by AR Foundation, which uses the ARCore subsystem to perform image tracking and detect the marker. Upon successful detection, the Home Scene is loaded, presenting the four food selection options. The user selects a food category and proceeds through the food customisation panels, where ingredient selections drive real-time price and calorie calculations. The final output is the order summary, which consolidates all user selections into a single display. The entire flow is managed by the ARScanController, SceneLoader, PanelManager, and respective food manager scripts.

4.2.1 Navigation Flow

The navigation flow of K-CustomAR is structured as a sequential scene-and-panel hierarchy. The application begins at the AR Scan Scene and progresses through the Home Scene to one of four food customisation scenes. Within each customisation scene, the PanelManager script controls sequential panel transitions without requiring additional scene loads. Figure 4.2 presents the navigation flow diagram of K-CustomAR.

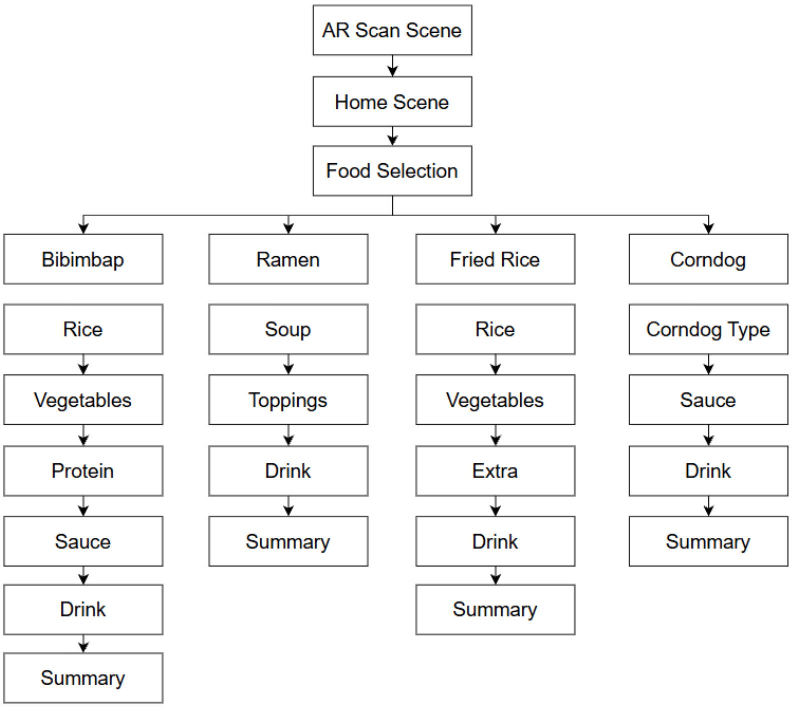

Figure 4.2 K-CustomAR Navigation Flow Diagram

The navigation flow proceeds as follows: the AR Scan Scene is the entry point where the user scans the menu marker; upon detection, the SceneLoader transitions to the Home Scene. From the Home Scene, the user selects Bibimbap, Ramen, Fried Rice, or Corndog. Within the selected scene, the user progresses through sequential customisation panels; and upon completion, the order summary is displayed. A return button on the order summary panel navigates the user back to the Home Scene to begin a new session.

4.3 User Interface Design

4.3.1 Navigation Design

K-CustomAR uses a tree-based navigation structure with the AR Scan Scene as the application entry point and the Home Scene as the central navigation hub. From the Home Scene, four branches lead to the individual food customisation scenes. The

Home Scene acts as the main navigation interface that allows users to select one of four Korean food modules. Figure 4.3 shows the Home Scene interface as implemented in the application.

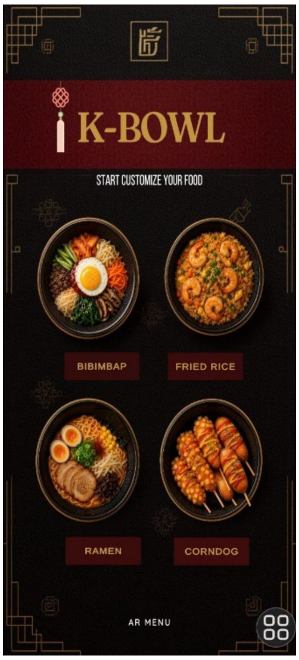

Figure 4.3 Home Scene Interface (Main Navigation Hub)

The navigation controls used throughout the application are tap buttons for scene transitions and panel progressions, toggle buttons for ingredient and sauce selections, a slider for rice portion selection in the Bibimbap module, and a confirmation button to display the order summary. Scene transitions are executed by the SceneLoader script using Unity's SceneManager.LoadScene() function, while panel transitions within each customisation scene are managed by the PanelManager script using SetActive().

4.3.2 Input Design

The application uses button-based inputs to simplify food customisation and reduce user interaction complexity. The following input types are implemented across the K-CustomAR application screens:

- Rice portion slider- used in the Bibimbap module to select Small, Medium, or Large portion size. The RiceSliderController script reads the slider position and updates the base price and calorie values accordingly.

- Tap buttons for food type selection- used in the Ramen module for soup base selection (Shoyu, Miso, Tonkotsu), in the Fried Rice module for rice type selection (Egg, Kimchi, Shrimp), and in the Corndog module for corndog type selection (Original, Potato, Mozzarella).
- Toggle buttons for ingredient and sauce selection- used across all four food modules to activate or deactivate individual ingredient options. The ToggleTopping script manages each toggle state and triggers a 3D model update and price/calorie recalculation on each toggle change.
- Drink selection buttons- used in all four modules via the DrinkManager script to select a drink accompaniment.

All inputs are validated implicitly through UI control constraints. No free-text input fields are used, eliminating invalid data entry. Figure 4.4 shows examples of the button-based input controls implemented across the customisation panels.

Figure 4.4 Button-Based Input Controls (Rice, Soup, Sauce, and Drink Selection Buttons)

4.3.3 Output Design

The system displays real-time outputs including food price, calorie information, and an order summary. All outputs are generated on an ad-hoc basis triggered by user interaction events. The following output types are produced:

- PriceText- a TextMesh Pro UI element displaying the real-time total price in Malaysian Ringgit, updated by the OrderManager after each ingredient selection change.

- CaloriesText- a TextMesh Pro UI element displaying the real-time total calorie count in kcal, updated simultaneously with the price display.
- SummaryText- a TextMesh Pro UI element in the order summary panel that consolidates all selected food item details, ingredient choices, total price, and total calorie count upon user confirmation.

Figure 4.5 shows the PriceText, CaloriesText, and SummaryText output elements as displayed in the application.

Figure 4.5 Output Display (PriceText, CaloriesText, and SummaryText in Application)

4.3.4 Data Management Design

K-CustomAR does not require an external database because all menu information used by the system is predefined and remains unchanged throughout operation. Food item details, ingredient options, pricing values, and calorie information are stored directly within the Unity application using C# scripts. This approach is appropriate for a prototype-level system where user accounts, online transactions, and persistent data storage are not required.

The application manages data through dedicated manager scripts, including OrderManager, RamenOrderManager, and FriedRiceOrderManager. These scripts store and process the relevant pricing and calorie values for each food category.

During customisation, the selected values are calculated dynamically to generate real-time updates of the total price and calorie information displayed to the user. This lightweight data management approach simplifies development while ensuring efficient performance and reliable operation throughout the customisation process.

4.3.5 Metaphors

The interface adopts a Korean-themed visual design using red, black, and gold colour schemes inspired by traditional Korean restaurant aesthetics. The warm red tones reference Korean chilli culture and the energy of Korean street food markets. The dark background conveys the modern Korean dining environment, while gold accent colours add a premium feel consistent with the food service context. The toggle switch interaction metaphor is used for ingredient selection, which is a widely understood mobile convention for the target demographic of users aged 18 to 35. Figure 4.6 shows the Home Scene UI demonstrating the Korean-themed visual design applied throughout the application.

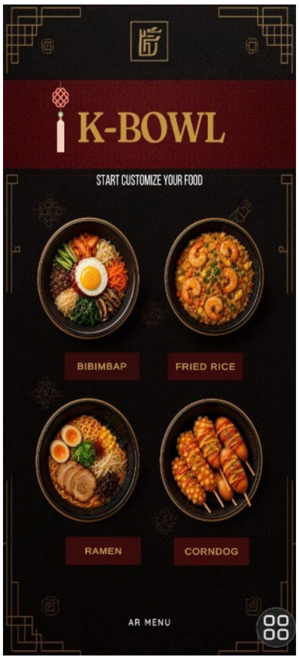

Figure 4.6 Home Scene UI (Korean-Themed Visual Design and Colour Scheme)

4.3.6 Template Design

A consistent UI template is applied across all four food customisation scenes to ensure a cohesive and predictable user experience. Each customisation scene follows the same panel structure: a top header displaying the food category name, a central 3D food model display area, a mid-section panel containing the current customisation step controls, and a bottom bar showing the live PriceText and CaloriesText values. The four customisation panels — Bibimbap Panel, Ramen Panel, Fried Rice Panel, and Corndog Panel — all follow this template structure. Figure 4.7 shows the panel template design applied across the K-CustomAR customisation scenes.

Figure 4.7 UI Panel Template Design (Bibimbap, Ramen, Fried Rice, and Corndog Panels)

4.3.7 Deployment to Mobile Device

K-CustomAR is deployed to Android devices through direct APK installation using Android Debug Bridge (ADB) via USB cable connection. The Unity project is compiled into an APK file using the Android Build Support module. The APK is transferred to the target device and installed directly without Google Play Store distribution. The target device must have Developer Mode, USB Debugging, and Install from Unknown Sources enabled to permit direct APK installation.

4.4 Media Creation

4.4.1 Production of Texts

TextMesh Pro (TMP) was used to display food names, prices, calorie information, and order summaries throughout the K-CustomAR application.

TextMesh Pro provides high-quality, resolution-independent text rendering using Signed Distance Field (SDF) technology, ensuring sharp text display across all Android screen densities. A clean sans-serif font is used for all UI labels, ingredient names, price displays, and calorie values, with a bold variant applied to section headings and food category titles. Font sizes range from 24 to 48 TMP units depending on the hierarchy level of the text element. Figure 4.8 shows the TextMesh Pro text components as configured in the Unity Inspector.

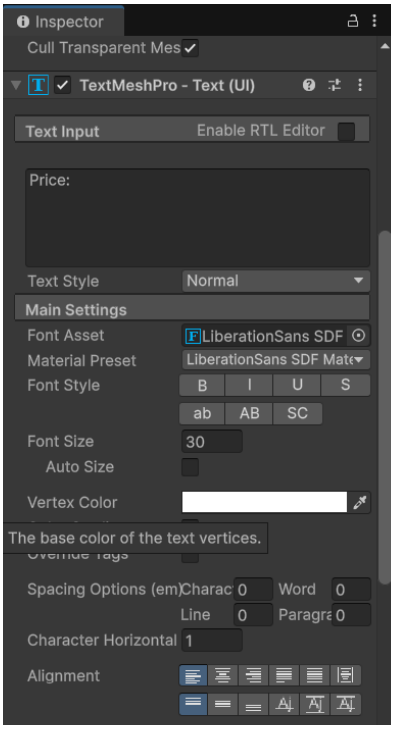

Figure 4.8 TextMesh Pro (TMP) Text Components in Unity Inspector

4.4.2 Production of Graphics

Food models and graphical assets were prepared and optimised using Blender before being imported into Unity. UI graphic assets including food category button images, panel backgrounds, and the AR marker image are produced as bitmap PNG files with transparency support at a minimum resolution of 1024 x 1024 pixels. The AR marker is designed as a high-contrast image to maximise detection reliability by the ARCore image tracking subsystem. All graphic assets are imported into Unity as Sprite (2D and UI) texture type and assigned to the respective UI Image components. Figure 4.9 shows the AR marker design and Figure 4.10 shows the 3D food models as displayed in the Unity Editor.

Figure 4.9 AR Marker Design Used for K-CustomAR Image Tracking

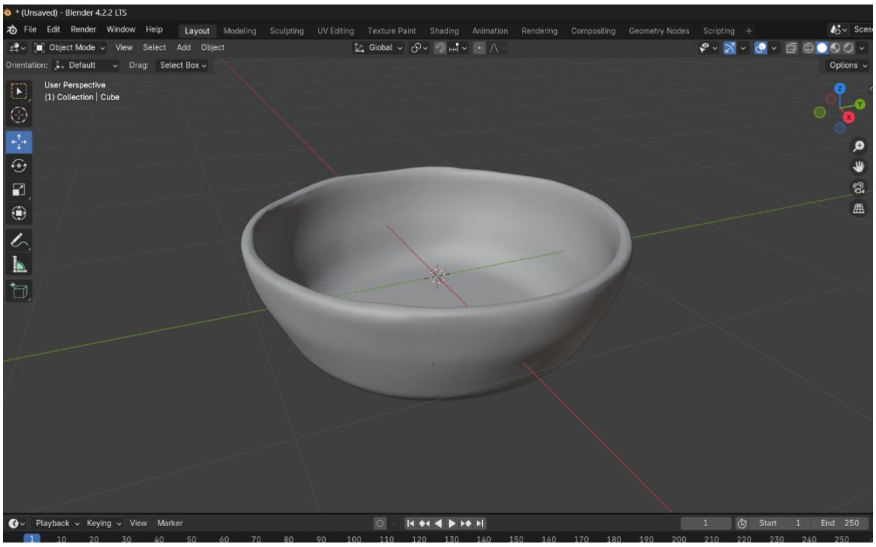

Figure 4.10 3D Food Models Prepared in Blender and Imported into Unity

4.4.3 Production of Audio

Button click sound effects were implemented using Unity AudioSource to provide user feedback during navigation and ingredient selection interactions. Audio clips are stored in WAV format at 44.1 kHz and compressed using the Vorbis codec within Unity to minimise APK size. Selected navigation buttons are configured with AudioSource components with Play On Awake disabled. Playback is triggered through OnClick() events to provide audio feedback during user interaction. Figure 4.11 shows the SoundManager configuration and Figure 4.12 shows the AudioSource setup in the Unity Inspector.

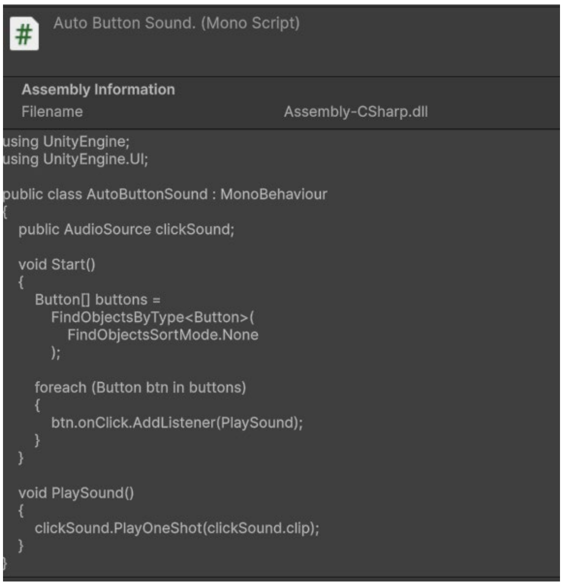

Figure 4.11 SoundManager Configuration in Unity

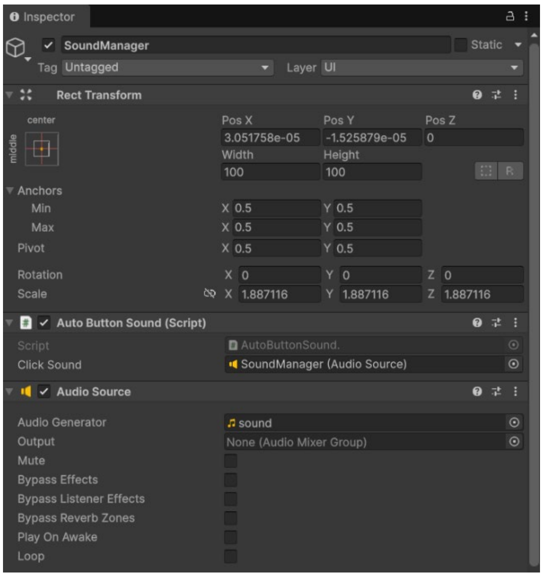

Figure 4.12 AudioSource Component Setup for Button Click Feedback

4.4.4 Production of Video

No pre-rendered video assets are used in K-CustomAR. The real-time three-dimensional rendering of food models within the Unity scene serves as the primary dynamic visual content, providing a more interactive and personalised visual experience than static video content. The food model display responds directly to the user's customisation selections in real time, driven by Unity's Universal Render Pipeline.

4.4.5 Production of Animation

4.4.5.1 Modelling, Mapping, Animation, Camera, Lighting and Texture

A pop-scale animation was implemented for ingredient components to provide visual feedback during food customisation. When a user selects an ingredient, the corresponding 3D model component is activated and a scaling animation is executed using the PopAnimation() coroutine. The animation enlarges the object briefly before returning it to its original scale, creating a smooth appearance effect that improves user interaction feedback. Figure 4.13 illustrates the scaling animation applied when a topping is selected. The selected ingredient briefly scales up before returning to its original size, providing visual confirmation that the topping has been successfully added to the customised food model.

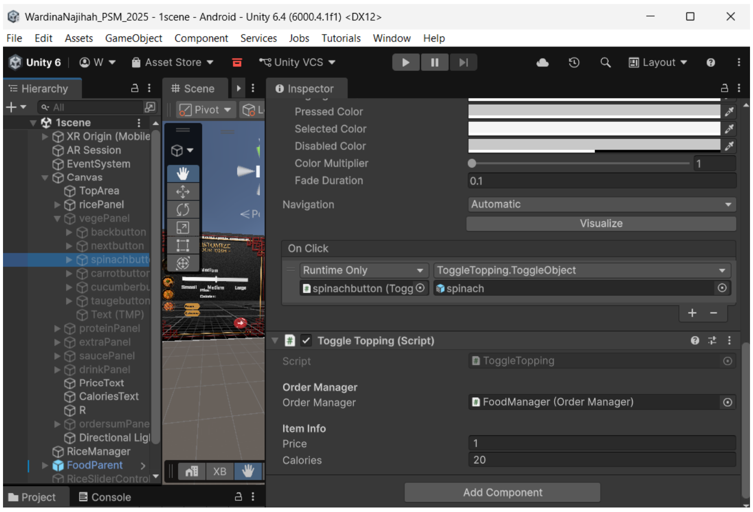

Figure 4.13 Toggle Topping Scaling Animation Applied to Ingredient 3D Model Components

Each food model is composed of individual mesh components prepared in Blender. UV unwrapping is applied to each component to support accurate texture mapping. PBR materials including base colour and roughness maps are applied. Scene lighting is managed by a directional light within each Unity customisation scene, supplemented by ambient lighting to ensure even illumination of the food model from the user's perspective.

4.4.5.2 Rendering Process

Food model components are exported from Blender in FBX format and imported into Unity. Materials are reassigned using the Universal Render Pipeline (URP) Lit shader for optimised mobile rendering. The URP is selected for its performance advantages on Android hardware. All model components are preloaded at scene initialisation and toggled using SetActive() during customisation interactions to avoid runtime loading delays.

4.5 Media Integration

All multimedia components created for K-CustomAR are integrated within the Unity 6 project through the Unity Asset Pipeline. This section presents the integration outcome for each food customisation module, supported by screenshots of the implemented application screens.

4.5.1 Bibimbap Module

The Bibimbap module is integrated across six sequential panels managed by PanelManager. The rice portion slider, vegetable toggle buttons, protein toggle buttons, sauce toggle buttons, and drink selection buttons are all connected to their respective script components through the Unity Inspector. The 3D Bibimbap model components are linked to the ToggleTopping and RiceSliderController scripts. Figures 4.14 to 4.19 show the implemented Bibimbap customisation panels.

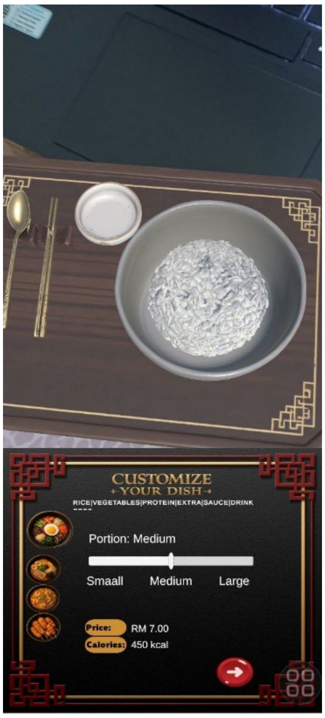

Figure 4.14 Bibimbap (Rice Portion Selection Panel)

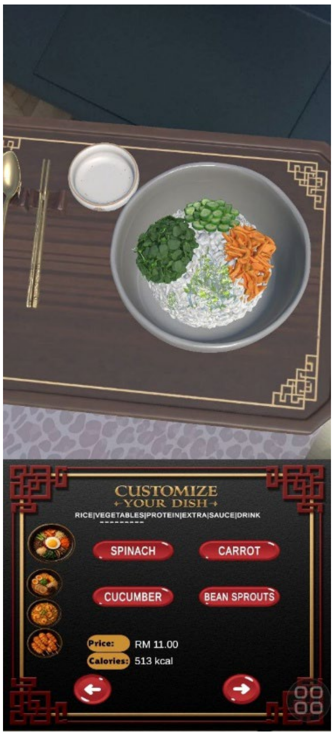

Figure 4.15 Bibimbap (Vegetable Selection Panel)

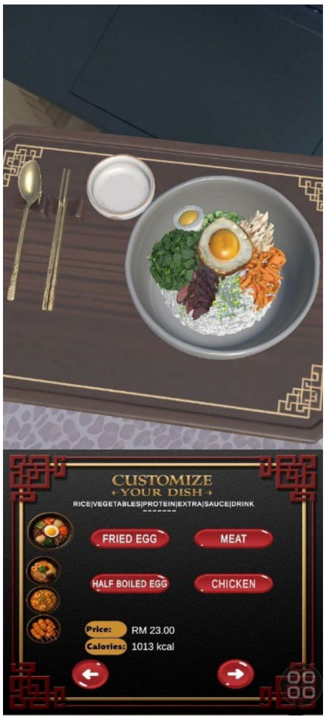

Figure 4.16 Bibimbap (Protein Selection Panel)

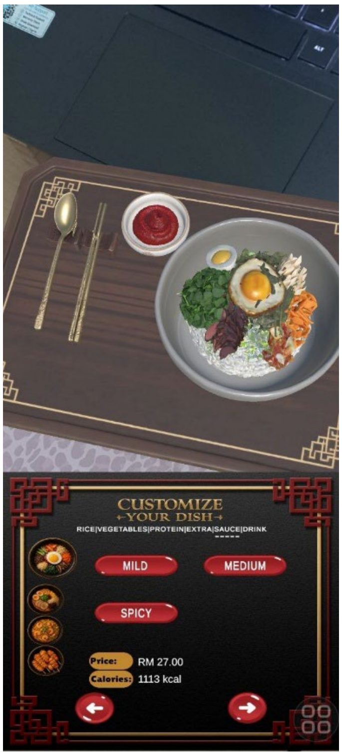

Figure 4.17 Bibimbap (Sauce Selection Panel)

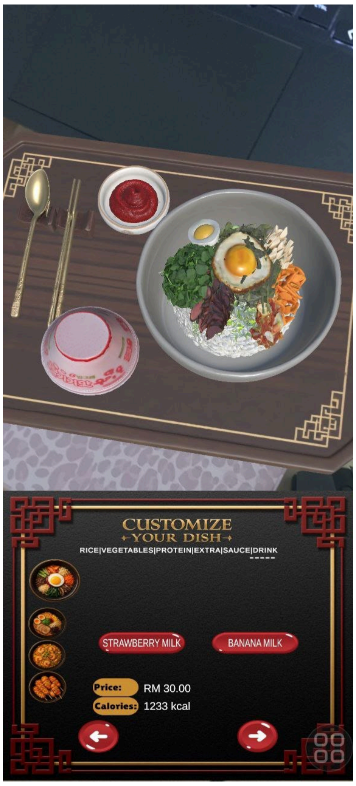

Figure 4.18 Bibimbap (Drink Selection Panel)

Figure 4.19 Bibimbap (Order Summary Panel)

4.5.2 Ramen Module

The Ramen module is integrated across four sequential panels. The soup base selection buttons are connected to the SoupManager script, and the topping toggle buttons are managed by ToggleTopping scripts. The DrinkManager handles drink selection. The 3D ramen model components are toggled based on the user's soup and topping selections. Figures 4.20 to 4.23 show the implemented Ramen customisation panels.

Figure 4.20 Ramen (Soup Base Selection Panel)

Figure 4.21 Ramen (Vegetables Selection Panel)

Figure 4.22 Ramen (Topping Selection Panel)

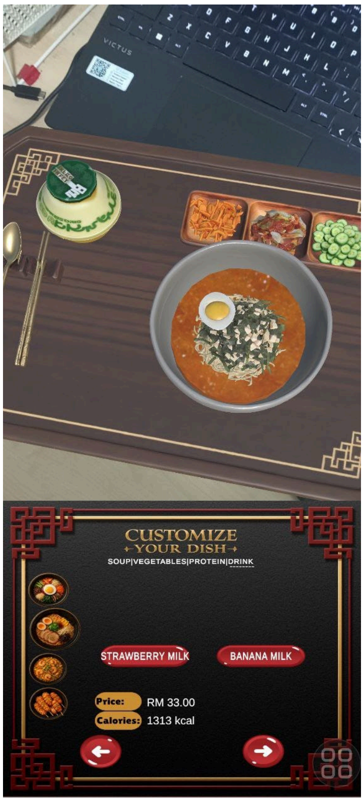

Figure 4.23 Ramen (Drink Selection Panel)

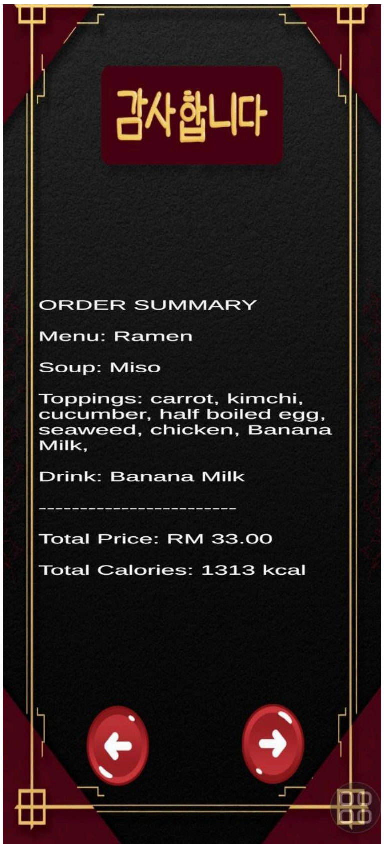

Figure 4.24 Ramen (Order Summary Panel)

4.5.3 Fried Rice Module

The Fried Rice module is integrated across five sequential panels. The FriedRiceManager handles rice type selection and the ToggleTopping scripts manage vegetable and extra ingredient selections. The DrinkManager handles drink selection. Figures 4.24 to 4.28 show the implemented Fried Rice customisation panels.

Figure 4.25 Fried Rice (Rice Type Selection Panel)

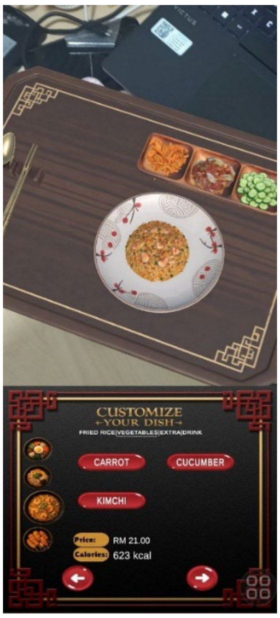

Figure 4.26 Fried Rice (Vegetable Selection Panel)

Figure 4.27 Fried Rice (Extra Ingredient Selection Panel)

Figure 4.28 Fried Rice (Drink Selection Panel)

Figure 4.29 Fried Rice (Order Summary Panel)

4.5.4 Corndog Module

The Corndog module is integrated across four sequential panels. The CorndogManager handles corndog type selection and sauce toggle selections. The DrinkManager handles drink selection. Figures 4.29 to 4.32 show the implemented Corndog customisation panels.

Figure 4.30 Corndog (Corndog Type Selection Panel)

Figure 4.31 Corndog (Sauce Selection Panel)

Figure 4.32 Corndog (Drink Selection Panel)

Figure 4.33 Corndog (Order Summary Panel)

4.6 Product Configuration Management

4.6.1 Configuration Environment Setup

The project configuration was managed using Unity 6 (version 6000.4.1f1) and Android Build Settings. The Unity Hub is used to manage the Unity installation and project access. Figure 4.33 shows the Unity Hub with the K-CustomAR project and the installed Unity 6 version. Figure 4.34 shows the Android Build Settings configured for the project.

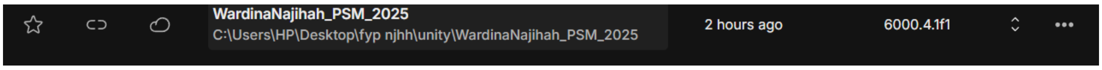

Figure 4.34 Unity Hub Showing Unity 6 (6000.4.1f1) Installation and K-CustomAR Project

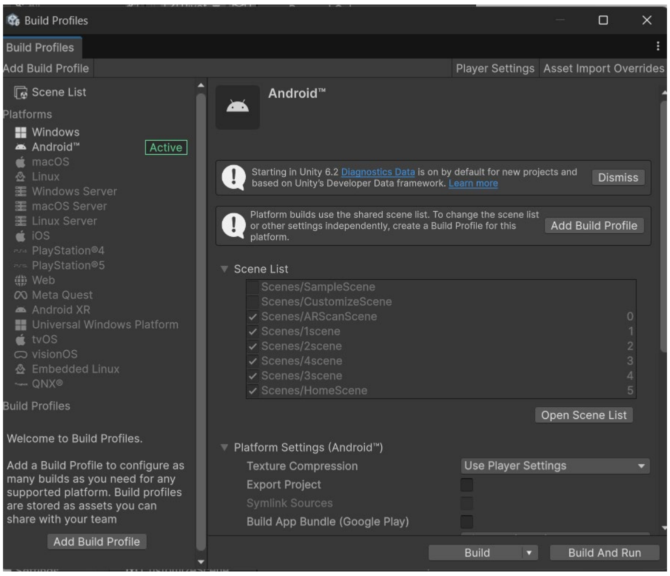

Figure 4.35 Android Build Settings Configured for K-CustomAR

The following packages and plugins are installed and configured through the Unity Package Manager: AR Foundation (version 6.x) for AR session management and image tracking, ARCore XR Plugin (version 6.x) for Android ARCore integration, TextMesh Pro (version 3.x) for text rendering, and Universal Render Pipeline (URP) for optimised Android rendering. The Android Player Settings are configured with Minimum API Level at Android 8.0 (API Level 26), Target API Level at Android 14 (API Level 34), and ARCore enabled under XR Plugin Management.

Additional logging scripts including PerformanceLogger and ARDetectionLogger were integrated into the Unity project to support future performance evaluation activities during the PSM 2 testing phase. The PerformanceLogger script records FPS, frame time, and memory usage to a CSV file at one-second intervals. The ARDetectionLogger script records the detection time for each AR image target from application launch to first detection. Figures 4.35 and 4.36 show these scripts as configured in the Unity project.

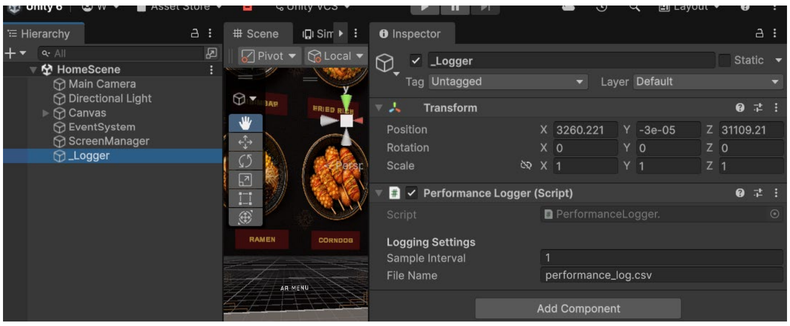

Figure 4.36 PerformanceLogger.cs Script Integrated in Unity Project

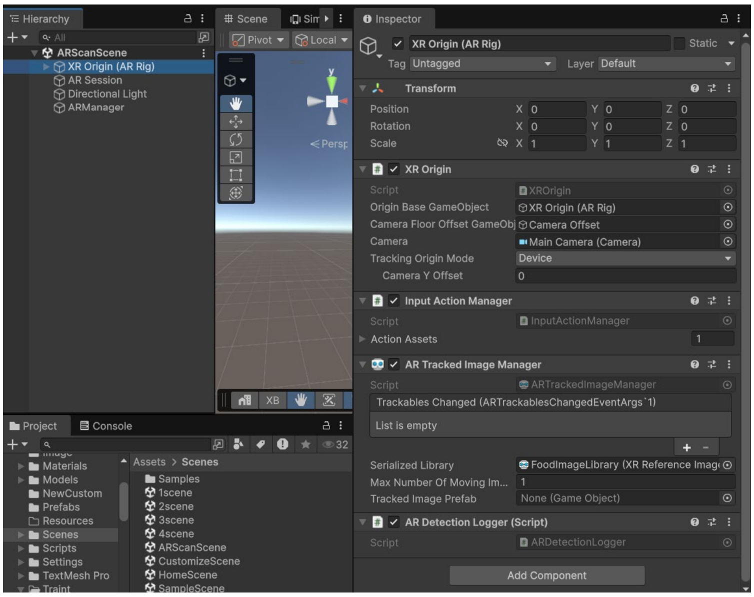

Figure 4.37 ARDetectionLogger.cs Script Integrated in Unity Project

4.6.2 Main Scripts Used in K-CustomAR

The development of K-CustomAR involves several scripts that support core system functionalities, including AR image tracking, scene navigation, food customization, interaction handling, order calculation, and performance monitoring. These scripts work together to ensure smooth system operation and user interaction throughout the application.

Table 4.1 Main Scripts Used in K-CustomAR

Script Name	Function
ARScanController	Detects image marker and loads the Home Scene.
SceneLoader	Handles navigation between scenes.
PanelManager, FriedRicePanelManager, CorndogPanelManager	Controls transitions between customization panels.
OrderManager,RamenOrderManager, FriedRiceOrderManager, CorndogOrderManager	Calculates price, calories, and generates order summaries.

DrinkManager,RamenDrinkManager, FriedRiceDrinkManager, CorndogDrinkManager	Handles drink selection and updates order information.
ToggleTopping, RamenToggleTopping, FriedRiceToggleTopping	Manages topping selection and updates price and calorie values.
RiceSliderController, RiceScaleOnly	Controls rice portion scaling and updates portion information.
SoupManager	Handles ramen soup selection.
FriedRiceManager	Handles fried rice type and extra ingredient selection.
CorndogManager	Handles corndog and sauce selection.
SauceAnimation	Controls sauce animation effects during customization.
TouchRotateZoom	Enables object rotation and zoom interaction in AR view.
LockPosition	Maintains food model position after placement.
AutoButtonSound	Plays button click sound effects.
PerformanceLogger	Records FPS and memory usage for performance evaluation.
ARDetectionLogger	Records AR image target detection time.

4.6.3 Version Control Procedure

Version control for K-CustomAR is managed through a structured release stage procedure. Project files are maintained with version-labelled backup folders at each release milestone following the naming convention KCustomAR_v[version]_[YYYYMMDD]. Table 4.1 presents the development milestones achieved throughout the implementation process.

Table 4.2 Version Control Stages for K-CustomAR

Version	Stage	Description
v0.1 Alpha	Alpha	AR marker detection and Home Scene navigation implemented. Basic SceneLoader functionality tested. Early ARCore integration verified on Android device.
v0.2 Alpha	Alpha	Bibimbap customisation module implemented with RiceSliderController and ToggleTopping scripts. 3D model components imported and SetActive() toggling verified.
v0.5 Beta	Beta	All four food customisation modules implemented. Real-time price and calorie calculation functional across all modules. Order summary panels integrated.
v0.8 RC1	Release Candidate 1	Full UI template applied across all scenes. Button click audio feedback integrated. PerformanceLogger.cs and ARDetectionLogger.cs integrated for performance logging support.
v0.9 RC2	Release Candidate 2	Bug fixes from RC1 applied. UI layout adjusted for different screen sizes. Preparation for testing across multiple Android device tiers.
v1.0 GM	Golden Master	Final stable build approved for PSM 2 user testing. All features verified and functional. APK packaged and ready for distribution to test participants.

4.7 Summary

This chapter discussed the implementation of K-CustomAR, including the system architecture, user interface design, media creation, integration process, and configuration management. The completed prototype successfully supports AR-based Korean food customisation with real-time price and calorie updates across four food modules: Bibimbap, Ramen, Fried Rice, and Corndog. The system is built using Unity 6 (version 6000.4.1f1), AR Foundation, ARCore, and C# scripting, and deployed as an Android APK. Performance logging scripts, PerformanceLogger and ARDetectionLogger, have been integrated in preparation for the technical evaluation phase. The next chapter will focus on system testing and evaluation, presenting the results of user testing conducted with a minimum of 30 participants and technical performance testing across three Android device tiers.

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